下载 Windows 驱动程序工具包 (WDK)

2025/07/01

WDK 用于开发、测试和部署 Windows 驱动程序。

- 可以在 Windows 7 及更高版本上安装并运行最新发布的 WDK。
- 还可以使用最新版本的 WDK 为 Windows 10、Windows Server 2016 及更高版本的客户端和服务器版本生成驱动程序。

> **重要**
>
> 从 2025 年 5 月开始，Microsoft不再发布旧版本的 Windows 驱动程序工具包。 对所有驱动程序开发工作使用最新版本的 WDK。 如果您要面向较旧版本的 Windows，请遵循博客文章**使用最新 Windows 驱动程序工具包（WDK）为早期操作系统版本构建驱动程序**中的指南。 如果最新的 WDK 不支持开发方案，请联系 **Microsoft WDK 反馈** 以获取帮助。

加入 Windows 预览体验计划 以获取 WDK Insider Preview 版本 。 有关 Windows Insider Preview 版本的安装说明，请参阅 安装 Windows 驱动程序工具包（WDK）的预览版。

WDK NuGet 包支持

从版本 10.0.26100.1 开始，WDK 作为 NuGet 包提供。 用户可以在 Visual Studio 中直接从 nuget.org 访问和使用这些包。 WDK NuGet 包提供了一种获取和更新 WDK 的便捷方法，它还管理 SDK 等依赖项，有助于保持驱动程序开发工具链的最新状态。 有关详细信息，请参阅使用 NuGet 安装最新的 WDK。

ARM64 支持

从 WDK 版本 10.0.26100.1 开始，WDK 现在支持在 ARM64 计算机上开发、测试和部署驱动程序。 WDK/EWDK 可以在 ARM64 硬件上安装并本地运行。 此外，之前支持的 ARM64 硬件上的 x86 KMDF/UMDF2 驱动程序模拟仍然可用。 此外，还支持从 ARM64 和 x64 主机调试驱动程序并将其部署到 ARM64 目标计算机。 在 ARM64 计算机上安装 WDK/EWDK 时，该过程会自动标识并安装所有必要的依赖项，包括生成工具、二进制文件和库。

步骤 1：安装 Visual Studio 2022

WDK 需要 Visual Studio。 有关 Visual Studio 系统要求的详细信息，请参阅 Visual Studio 2022 系统要求。

以下版本的 Visual Studio 2022 支持针对此发行版进行驱动程序开发：

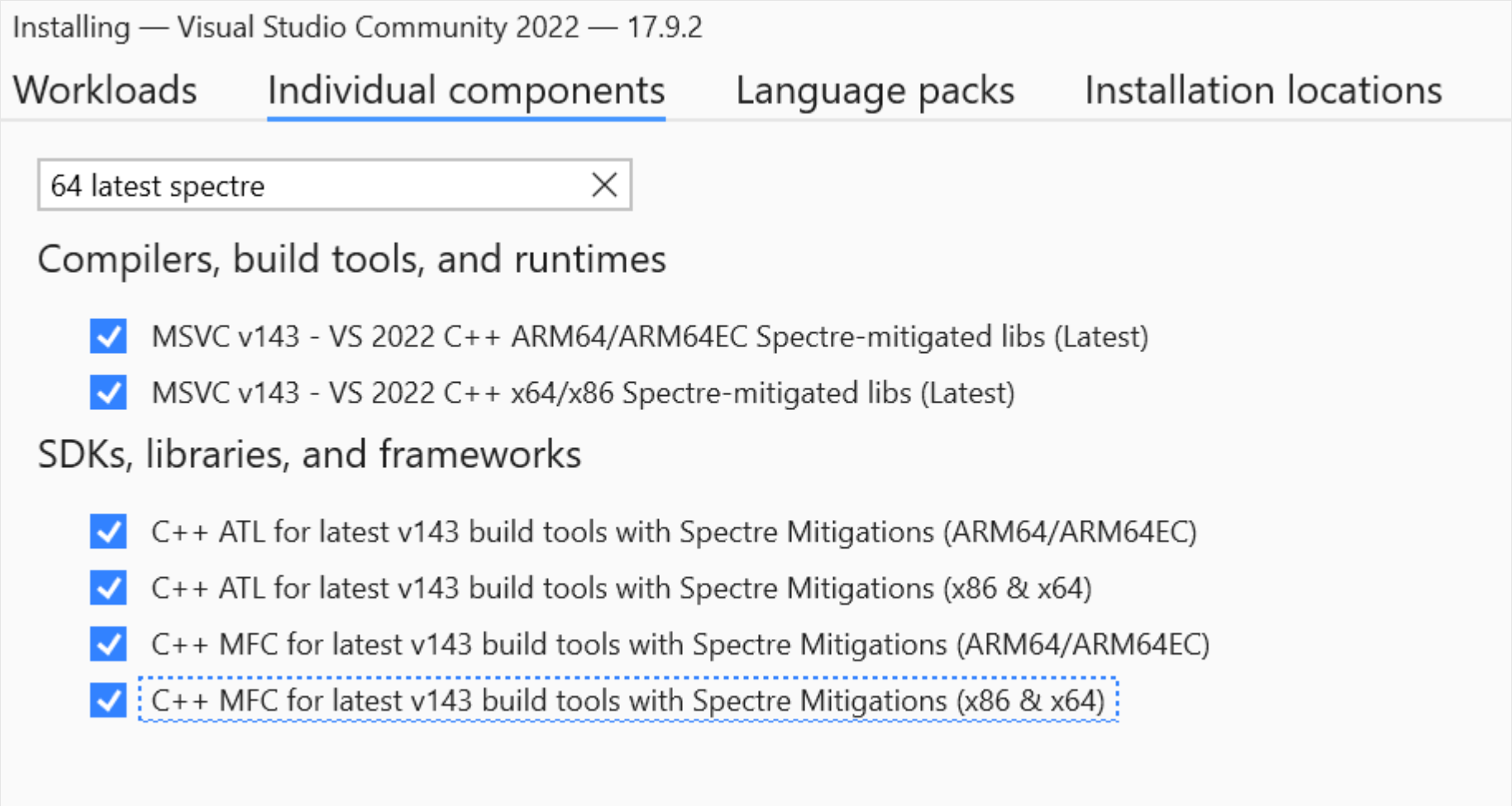
- 下载 Visual Studio Community 2022
- 下载 Visual Studio Professional 2022
- 下载 Visual Studio Enterprise 2022

安装 Visual Studio 2022 时，选择**使用 C++ 进行桌面开发**工作负荷，然后在“单独组件”下添加：

- MSVC v143 - VS 2022 C++ ARM64/ARM64EC Spectre 缓解库（最新版本）
- MSVC v143 - VS 2022 C++ x64/x86 Spectre 缓解库（最新版本）
- 带有 Spectre 缓解库的适用于最新 v143 生成工具的 C++ ATL (ARM64/ARM64EC)
- 带有 Spectre 缓解库的适用于最新 v143 生成工具的 C++ ATL (x86 & x64)
- 带有 Spectre 缓解库的适用于最新 v143 生成工具的 C++ MFC (ARM64/ARM64EC)
- 适用于最新 v143 生成工具且带有 Spectre 漏洞缓解措施的 C++ MFC (x86 & x64)
- Windows 驱动程序工具包 (WDK)

提示

使用搜索框查找“64 latest spectre”（在英文安装中）或“64 最新”（在非英文安装中）以快速查看这些组件。



备注

所选的**C++ 桌面开发**工作负载不会安装 Windows SDK 10.0.26100.1。

步骤 2：安装 SDK

安装 Visual Studio 不会下载最新的 SDK 版本，请使用以下链接进行安装

- 下载最新的 Windows SDK

所提供的 SDK 和 WDK 链接具有匹配的版本号，这对套件的协同工作始终必不可少。 如果决定为不同的 Windows 版本安装自己的 SDK/WDK 对，请确保版本号保持一致。 有关详细信息，请参阅 工具包版本控制。

步骤 3：安装 WDK

- 下载最新的 WDK

从版本 17.11.0 开始，WDK VSIX 作为单个组件包含在 Visual Studio 中。 安装 WDK 之前，安装程序会检查是否已安装兼容版本的 VSIX。 如果未找到 WDK VSIX，系统会提示用户安装它。 若要安装 WDK VSIX，请启动 Visual Studio 安装程序，选择“**修改**”，导航到“**各个组件**”选项卡，添加 **Windows 驱动程序工具包**，然后选择“**修改**”。

Windows Driver Kit - Windows 10.0.26100.2161

Important Information!

The Windows Driver Kit (WDK) requires the "Windows Driver Kit" visual studio individual component to be installed with Visual Studio 2022. At least one Visual Studio instance does not comply with this prerequisite and will not have a build environment for drivers when installation completes.

Please launch the Visual Studio Installer, and modify your installation to include the "Windows Driver Kit" individual component.

If you want to install the WDK without having installed the "Windows Driver Kit" individual component or to other instances with the component already installed, please click Next.

Back | Next | Close

提示

如果在 Visual Studio 中找不到驱动程序项目模板，则表示 WDK Visual Studio 扩展未正确安装。 若要解决此问题，请启动 Visual Studio Installer，选择 修改，在“单个组件”选项卡中添加 Windows 驱动程序工具包，然后选择“修改”。

企业版 WDK (EWDK)

除了下载 Visual Studio、SDK 和 WDK 之外，还可以下载 EWDK，这是一个用于生成驱动程序的独立命令行环境。 其中包括 Visual Studio 生成工具、SDK 和 WDK。

EWDK 的最新公开版本包含 Visual Studio 2022 Build Tools 17.11.4 和 MSVC 工具集 v14.41

EWDK 还需要 .NET Framework 版本 4.7.2。 有关 .NET Framework 的其他要求的详细信息，请参阅 .NET Framework 系统要求。

- 使用 Visual Studio 生成工具下载 EWDK

下载 ISO 后，使用以下步骤设置生成环境：

1. 从驱动器卷挂载 EWDK ISO。 目前不支持网络共享路径。
2. 运行 *LaunchBuildEnv.cmd*。
3. 在步骤 2 中创建的环境中，键入“SetupVSEnv”，然后按“Enter”。
4. 使用完整的文件路径，从同一环境中启动 devenv.exe。 例如：`"C:\Program Files\Microsoft Visual Studio\2022\%Community|Professional|Enterprise%\Common7\IDE\devenv.exe"`
5. 使用完生成环境后，可能想弹出 ISO。

可以选择使用 Visual Studio 界面和 EWDK 中提供的构建工具。 若要使用 Visual Studio 接口，请确保 Visual Studio 主版本与 EWDK 中的Visual Studio 生成工具版本匹配。 例如，Visual Studio 2022 适用于包含 VS17.X 生成工具的 EWDK。 有关 Visual Studio 2022 版本号的列表，请参阅 Visual Studio 2022 版本。

套件版本控制

完整的套件内部版本字符串包括最后两个部分，即内部版本号和 QFE（快速修复工程）值。 例如，10.0.22621.2428 的内部版本号为 22621，QFE 值为 2428。

要生成驱动程序，SDK 安装的*内部版本号必须与* WDK 安装的*内部版本号*一致。 除非驱动程序使用的功能仅在后续 QFE 附带的标头中可用，否则 QFE 值无需匹配。

查看本地安装的工具包完整生成字符串的快速方法是转到 Windows 设置（Win+I），导航到**应用**，然后选择**已安装的应用**，在**搜索**框中输入 `kit`。 完整的构建字符串显示在套件名称右边。 如果导航到 `C:\Program Files (x86)\Windows Kits\10\Include`，则显示的 QFE 将硬编码为 `.0`。 因此，目录名称不是检查 QFE 标识符的可靠方法。 安装套件时，新安装将替换以前任何具有相同版本号的现有安装。 在通过**使用 C++ 的桌面开发**工作负荷安装 Visual Studio 时，如果安装负载包括 Windows SDK，则右侧的摘要窗格还会显示 QFE 的硬编码 `.0`。

Windows 驱动程序示例

通过以下方式之一下载驱动程序示例：

- 转到 GitHub 上的**驱动程序示例页，选择“克隆”或“下载”，然后选择“下载 ZIP”**。
- 下载适用于 Visual Studio 的 GitHub 扩展，然后连接到 GitHub 存储库。
- 浏览 Microsoft 示例门户中的驱动程序示例。

相关下载

- 下载 WDK Insider Preview
- 下载以前版本的 WDK
- 下载 Windows 评估和部署工具包 (Windows ADK)
- 下载 Windows HLK
- 下载 Windows 调试工具 (WinDbg)
- 下载 Windows 符号包

另请参阅

- Windows 11 硬件要求
- 使用 WinGet 安装 WDK
- 了解驱动程序开发的新增功能
- 查看已知问题

https://learn.microsoft.com/zh-cn/windows-hardware/drivers/download-the-wdk

Download the Windows Driver Kit (WDK)

07/01/2025

The WDK is used to develop, test, and deploy drivers for Windows.

- You can install and run the latest released WDK on Windows 7 and later.
- You can also use the latest release WDK to build drivers for Windows 10, Windows Server 2016, and later client and server versions.

> **Important**
>
> Starting in May 2025, Microsoft no longer publishes older versions of Windows Drivers Kit. Use the latest release of the WDK for all driver development efforts. If you're targeting older versions of Windows, follow the guidance in the Building Drivers for Previous OS Releases Using the Latest Windows Driver Kit (WDK) blog post. If your development scenario is not supported by the latest WDK, contact Microsoft WDK Feedback for assistance.

Join the Windows Insider Program to get WDK Insider Preview builds. For installation instructions for Windows Insider Preview builds, see Installing preview versions of the Windows Driver Kit (WDK).

WDK NuGet package support

WDK is available as a NuGet package starting from version 10.0.26100.1. Users can access and use these packages directly from nuget.org within Visual Studio. WDK NuGet package provides a convenient way to acquire and update the WDK, it also manages dependencies such as the SDK, helping to keep the driver development toolchain current. For more information, see Install the latest WDK using NuGet.

ARM64 support

Beginning in WDK version 10.0.26100.1, the WDK now supports the development, testing, and deployment of drivers on ARM64 machines. The WDK/EWDK can be installed and run natively on ARM64 hardware. Additionally, the previously supported emulation of x86 KMDF/UMDF2 drivers on ARM64 hardware is still available. Also, debugging and deploying drivers to an ARM64 target machine is now supported from both ARM64 and x64 host machines. When you install the WDK/EWDK on ARM64 machines, the process automatically identifies and installs all necessary dependencies, including build tools, binaries, and libraries.

Step 1: Install Visual Studio 2022

The WDK requires Visual Studio. For more information about system requirements for Visual Studio, see Visual Studio 2022 System Requirements.

The following editions of Visual Studio 2022 support driver development for this release:

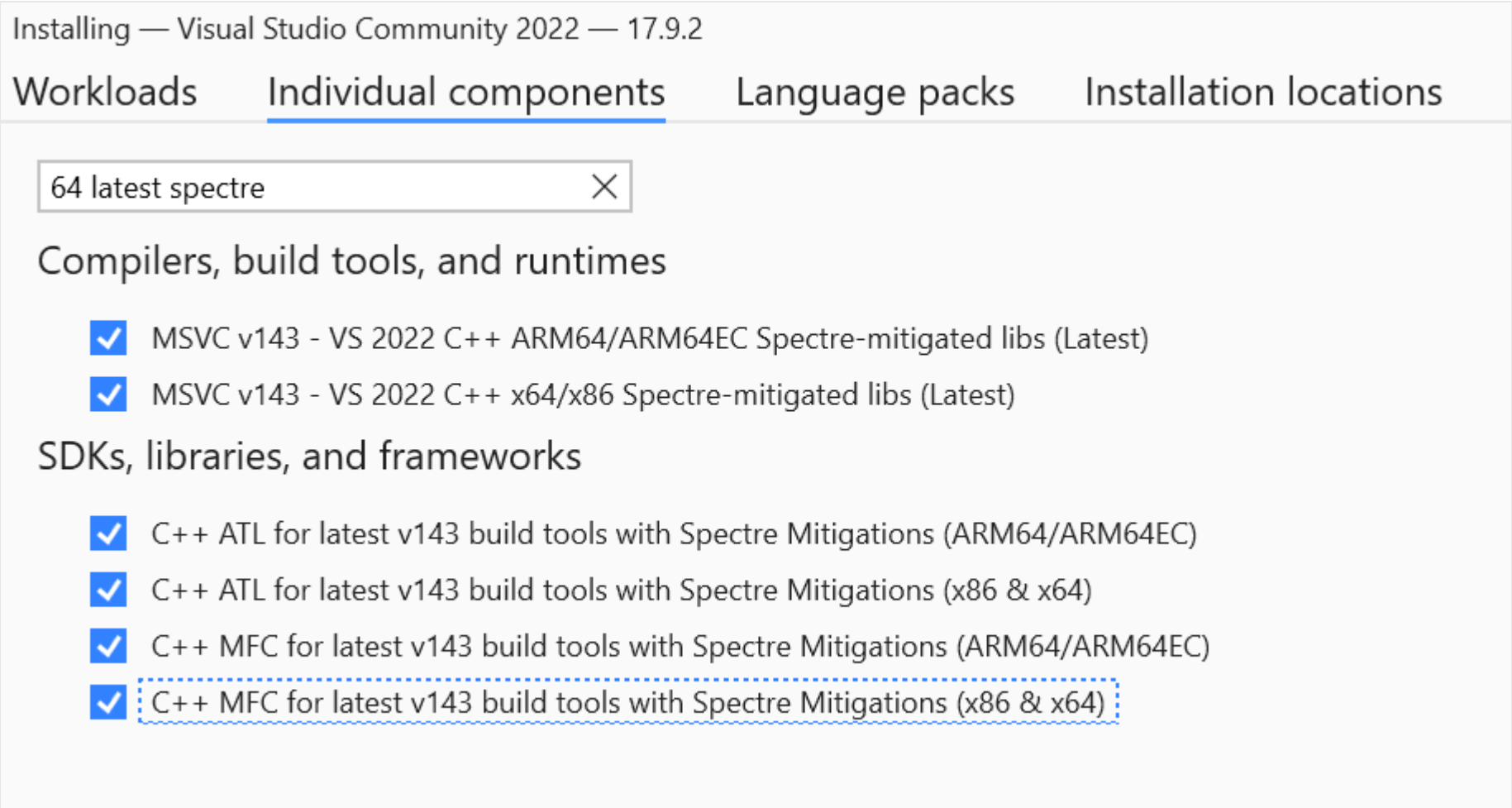
- Download Visual Studio Community 2022
- Download Visual Studio Professional 2022
- Download Visual Studio Enterprise 2022

When you install Visual Studio 2022, select the **Desktop development with C++** workload, then under Individual Components add:

- MSVC v143 - VS 2022 C++ ARM64/ARM64EC Spectre-mitigated libs (Latest)
- MSVC v143 - VS 2022 C++ x64/x86 Spectre-mitigated libs (Latest)
- C++ ATL for latest v143 build tools with Spectre Mitigations (ARM64/ARM64EC)
- C++ ATL for latest v143 build tools with Spectre Mitigations (x86 & x64)
- C++ MFC for latest v143 build tools with Spectre Mitigations (ARM64/ARM64EC)
- C++ MFC for latest v143 build tools with Spectre Mitigations (x86 & x64)
- Windows Driver Kit

Tip

Use the Search box to look for "64 latest spectre" (in English installations) or "64 latest" (in non-English installations) to quickly see these components.



Note

The **Desktop development with C++** workload selected does not install the Windows SDK 10.0.26100.1.

Step 2: Install SDK

Installing Visual Studio doesn't download the latest SDK version, use the following link to install

- Download the latest Windows SDK

The provided links for the SDK and the WDK have matching build numbers, which is always required for the kits to work together. If you decide to install your own SDK/WDK pair, perhaps for a different Windows version, ensure that the build numbers match. For more information, see Kit versioning.

Step 3: Install WDK

- Download the latest WDK

Starting with version 17.11.0, the WDK VSIX is included as an individual component in Visual Studio. Before installing the WDK, the installer checks if a compatible version of the VSIX is already installed. If the WDK VSIX is not found, users will be prompted to install it. To install the WDK VSIX, launch the Visual Studio Installer, select **Modify**, navigate to the **Individual Components** tab, add **Windows Driver Kit**, and then select **Modify** again.

Windows Driver Kit - Windows 10.0.26100.2161

Important Information!

The Windows Driver Kit (WDK) requires the "Windows Driver Kit" visual studio individual component to be installed with Visual Studio 2022. At least one Visual Studio instance does not comply with this prerequisite and will not have a build environment for drivers when installation completes.

Please launch the Visual Studio Installer, and modify your installation to include the "Windows Driver Kit" individual component.

If you want to install the WDK without having installed the "Windows Driver Kit" individual component or to other instances with the component already installed, please click Next.

Back | Next | Close

> **Tip**
>
> If you can't find driver project templates in Visual Studio, the WDK Visual Studio extension didn't install properly. To resolve this, launch Visual Studio Installer, select **Modify**, add **Windows Driver Kit** in the **Individual Component** tab, and select **Modify**.

Enterprise WDK (EWDK)

As an alternative to downloading Visual Studio, the SDK, and the WDK, you can download the EWDK, which is a standalone, self-contained command-line environment for building drivers. It includes Visual Studio Build Tools, the SDK, and the WDK.

The latest public version of the EWDK contains Visual Studio 2022 Build Tools 17.11.4 and MSVC toolset v14.41

The EWDK also requires the .NET Framework version 4.7.2. For more information about other requirements for the .NET Framework, see .NET Framework system requirements.

- Download EWDK with Visual Studio Build Tools

After you download the ISO, use these steps to set up your build environment:

1. Mount the EWDK ISO from a drive volume. Network share paths aren't currently supported.
2. Run *LaunchBuildEnv.cmd*.
3. In the environment created in step 2, type **SetupVSEnv**, and then press **Enter**.
4. Launch devenv.exe from the same environment, using the full file path. For example: `"C:\Program Files\Microsoft Visual Studio\2022\%Community|Professional|Enterprise%\Common7\IDE\devenv.exe"`
5. When you're done with the build environment, you might want to eject the ISO.

You can optionally use the Visual Studio interface with the build tools provided in the EWDK. To use the Visual Studio interface, ensure that the Visual Studio major version matches the version of the Visual Studio Build Tools in the EWDK. For example, Visual Studio 2022 works with the EWDK that contains VS17.X build tools. For a list of Visual Studio 2022 version numbers, see Visual Studio 2022 Releases.

Kit versioning

A full kit build string includes as its last two components, the build number and a QFE (Quick Fix Engineering) value. For example, 10.0.22621.2428 has a build number of 22621, and a QFE value of 2428.

To build a driver, the *build number* of your SDK installation must match the *build number* of your WDK installation. The QFE values don't need to match unless your driver uses functionality that is only available in the headers included with a later QFE.

A quick way to see the full build string for locally installed kits is to go to Windows settings (Win+I), navigate to **Apps**, then **Installed apps**, and in the **Search** box type `kit`. The full build string appears to the right of the kit name. If you navigate to `C:\Program Files (x86)\Windows Kits\10\Include`, the QFE shown is hardcoded to `.0`. So, the directory name isn't a reliable way to check your QFE identifier. When you install a kit, the new installation replaces any previously existing installation of the same build number. When you install Visual Studio with the **Desktop development with C++** workload, if the installation payload includes the Windows SDK, the right-hand Summary pane also shows a hardcoded `.0` for QFE.

Driver samples for Windows

Download the driver samples in one of these ways:

- Go to the driver samples page on GitHub, select **Clone or download**, and then select **Download ZIP**.
- Download the GitHub Extension for Visual Studio, and then connect to the GitHub repositories.
- Browse the driver samples on the Microsoft Samples portal.

Related downloads

- Download the WDK Insider Preview
- Download previous versions of the WDK
- Download the Windows Assessment and Deployment Kit (Windows ADK)
- Download the Windows HLK
- Download the Windows Debugging Tools (WinDbg)
- Download Windows Symbol Packages

See also

- Windows 11 hardware requirements
- Install the WDK using WinGet
- Learn what's new in driver development
- Review known issues